LECTURE PRESENTATION

System Analysis & Design

System Interface Design, Navigation Design, Input Design and Output Design

Presented by Team 4

INTRODUCTION

Learning Overview

Six parts — from the principles behind good interfaces to applying them in a complete real-world system.

01 **UI Design Principles**
Foundations of usable, aesthetic, and low-effort interfaces

02 **UI Design Process**
From use scenarios to iterative prototyping and evaluation

03 **Navigation Design**
Principles, controls, components, menus, and messaging

04 **Input Design & Validation**
Source data capture, controls, and multi-tier validation

05 **Output Design & Case Study**
Reports, dashboards, and the end-to-end Tune Source journey

06 **Summary & Discussion**
Key takeaways, seminar activity, and academic discussion

Introduction to UI Design

The user interface connects human users to underlying software capabilities, shaping interaction, satisfaction, and operational efficiency.

Purpose of UI Design

Bridges user intent and software capability; translates goals into intuitive action.

User Interaction

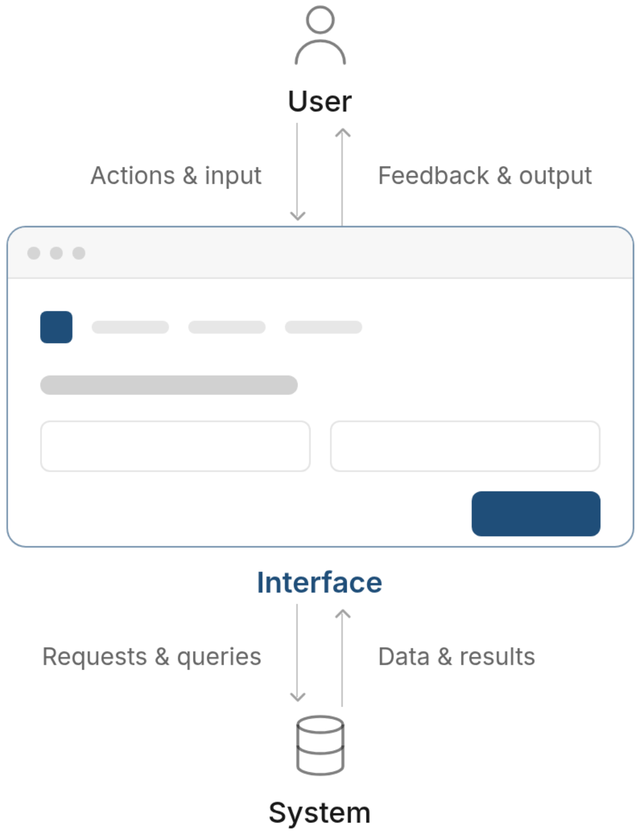
Two-way dialogue: users issue commands, the system responds with immediate feedback.

User Satisfaction

Fosters trust, ease, and comfort while minimizing cognitive strain and friction.

System Efficiency

Maximizes task speed, throughput, and accuracy with minimal human error.



UI Design Principles

Eight foundational principles that govern the usability, learnability, and longevity of system interfaces.

01

Layout

Organise every screen into predictable, consistent areas (nav, work, status).

02

Content Awareness

Ensure users always know where they are, what they see, and what happened.

03

Aesthetics

Pleasing, functional visual design with balanced whitespace and clear hierarchy.

04

User Experience

Intuitive, satisfying, and easy to learn for novices while efficient for experts.

05

Consistency

Identical behaviors, terms, icons, and visual elements across the system.

06

Minimize User Effort

Reduce cognitive load, keystrokes, and clicks (the Three-Clicks Rule).

07

Accessibility

Ensure equal access across abilities, assistive technologies, and devices.

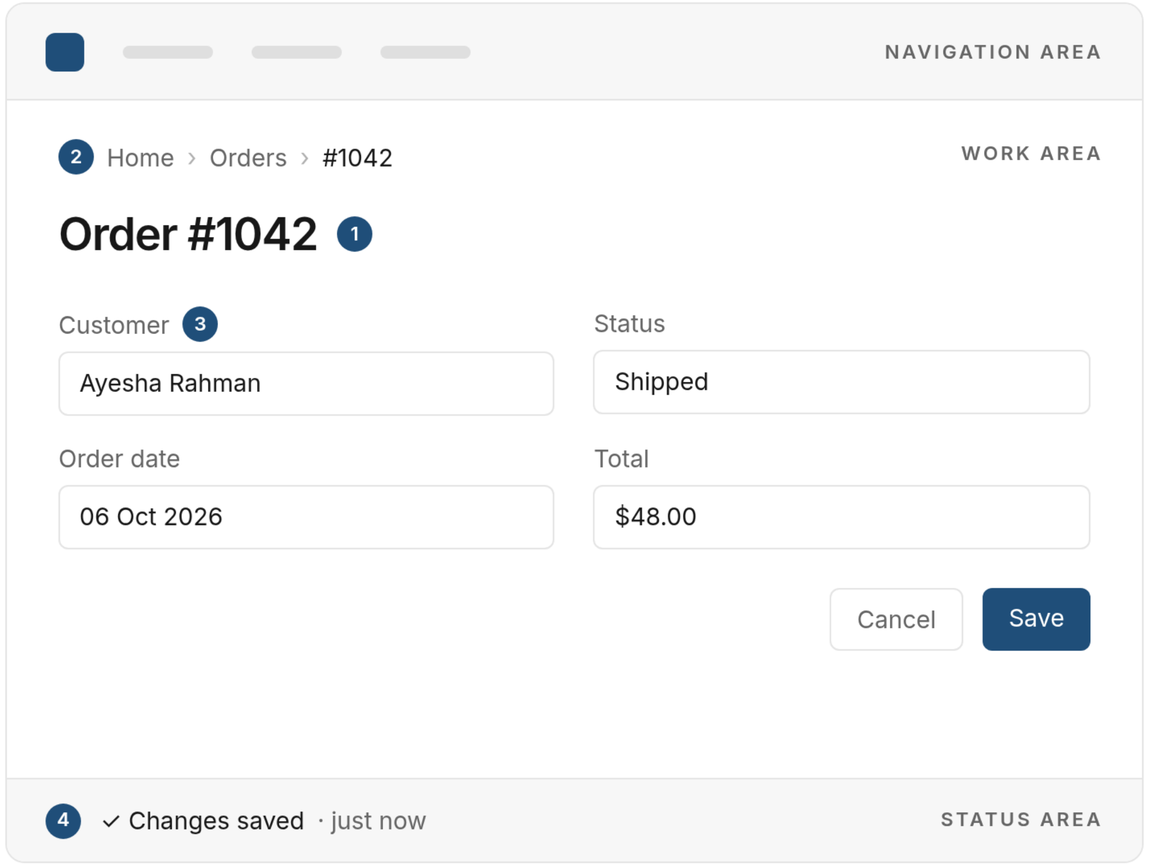
08

Scalability

Accommodate feature expansion, large datasets, and evolving user requirements.

Layout & Content Awareness

Standardized screen zones paired with explicit contextual signposts keep users oriented.



Layout

Divide each screen into three functional zones — navigation, work, and status — and keep their positions uniform across every screen in the system.

Content awareness

Users should never feel lost or uncertain about system state.

1. **Page titles** — Explicitly name the current screen and its purpose
2. **Breadcrumbs** — Show the hierarchical path — and provide a one-click way back
3. **Labels** — Unambiguously describe every input field and content section
4. **Feedback** — Immediately confirm system actions and status changes

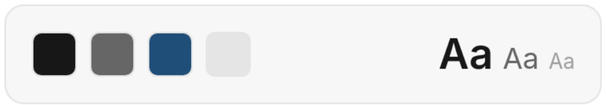
Aesthetics, UX & Consistency

Visual elegance, operational experience, and system-wide predictability work together to eliminate friction.

VISUAL DESIGN — LOOKS

Aesthetics

How the interface appears visually.



Purposeful Colors

High-contrast palette; supports meaning without being the sole cue.

Typography Scale

Clear hierarchy across headings, body, and labels with readable fonts.

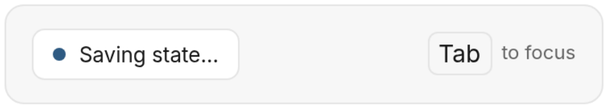
Whitespace & Balance

Chunks related content; provides breathing room to prevent clutter.

INTERACTION DESIGN — FEELS

User Experience

How the interface feels during operation.



Responsiveness

Instant feedback on every action; layouts adapt across screen sizes.

Learnability & Ease

Intuitive mental models for quick learning and low cognitive fatigue.

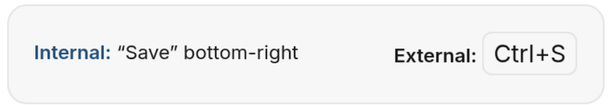
Accessibility & Safety

Full keyboard support, clear contrast, and tolerant error recovery.

SYSTEM STANDARDS — BEHAVES

Consistency

How the interface behaves predictably.



Internal Consistency

Same terms, layout zones, and button placements across every screen.

External Conventions

Adheres to common platform norms (e.g. Ctrl+S, logo links home).

Muscle Memory

Predictable interactions prevent user hesitation and relearning.

Minimize User Effort

Anticipate user intent, eliminate unnecessary keystrokes, and enforce the Three-Clicks Rule.

EFFORT REDUCTION 01

Autofill

Pre-fills saved profile or browser data.

SHIPPING ADDRESS Auto-filled

Ayesha Rahman

42 Park Road, Block B

Benefit: Eliminates repetitive manual typing.

EFFORT REDUCTION 02

Predictive Text

Suggests matching queries in real time.

LIVE SEARCH FIELD

Wire

Wireless Headphones

Wireless Charger

Wired Earbuds

Benefit: Prevents typos and speeds search.

EFFORT REDUCTION 03

Shortcuts

Keyboard accelerators for power users.

QUICK ACTION BAR

Quick Save Ctrl + S

Global Search /

Undo Action Ctrl + Z

Benefit: Bypasses mouse navigation for speed.

EFFORT REDUCTION 04

Remember Me

Persists session tokens across return visits.

AUTHENTICATION STATE

Keep me signed in

Secure token Active · 30 Days

Benefit: Eliminates repeated login friction.

3×

The Three-Clicks Rule

Users should be able to navigate to any function, document, or piece of information in no more than three mouse clicks or taps from the start screen.

Standard HCI Benchmark

UI Design Process Overview

A cyclical five-stage engineering process, with prototyping and evaluation repeated until usability criteria are met.



Scenario Development

Capture real-world user goals, workflows, and task sequences

Structure Design

Define screens and navigational relationships with an ISD

Standards Design

Establish interface metaphors, terminology, icons, and templates

Prototyping

Build iterative mockups ranging from paper sketches to code

Evaluation

Conduct heuristic reviews and user tests to validate the experience

Scenario Development & Structure Design

How task-based use scenarios directly translate into the system’s structural screen hierarchy and navigation flow.

STEP 01 · TASK ANALYSIS — SCENARIOS

Use Scenario Development

Outlines the exact steps a user follows to complete a specific goal.

SAMPLE TASK FLOW — Album Purchase

1. Search catalog & audition track
2. Select media format & add to cart
3. Confirm shipping & submit payment

User Goals

Target outcome driving the session.

Workflows

Step-by-step path from start to completion.

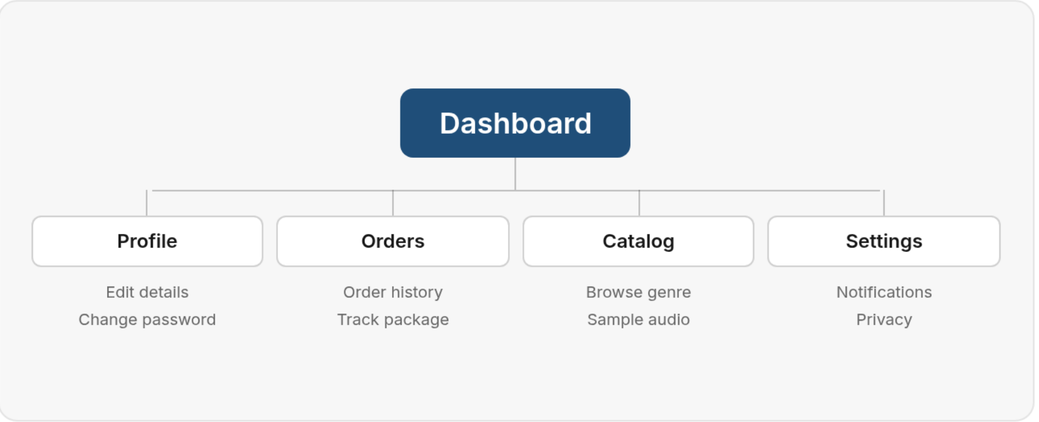
Edge Cases & Errors

Graceful handling of exceptions (e.g. stockout).

STEP 02 · ARCHITECTURAL MAP — STRUCTURE (ISD)

Interface Structure Design

An Interface Structure Diagram (ISD) maps every screen and navigational route.



Key Integration: Scenarios identify what screens are needed; the ISD defines how users transition between them without getting trapped.

Standards Design & Prototyping

Interface standards establish design conventions, while iterative prototyping moves concepts from sketch to functional code.

INTERFACE STANDARDS

Agreed-upon design rules and conventions shared by every screen in the system.

Interface Metaphors

Real-world concepts guiding user intuition — shopping cart, desktop, folder, media player.

Objects, Actions & Icons

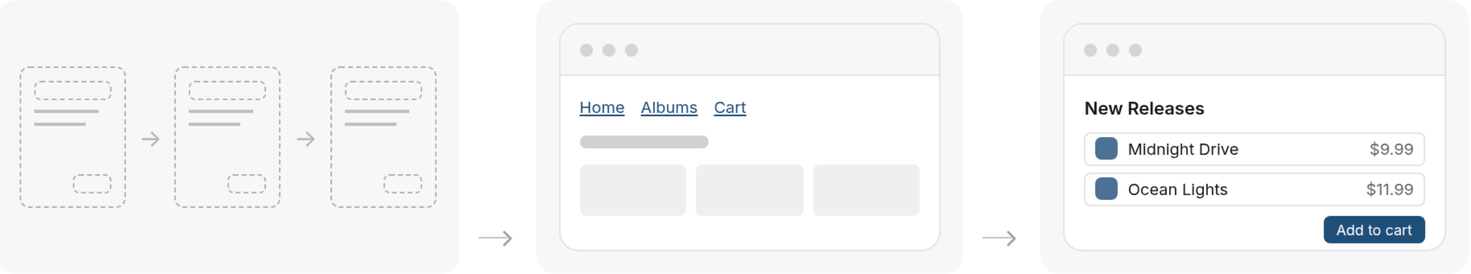
Standardized terminology and iconography across every module (e.g., “Archive” vs “Delete”).

Screen Templates

Common grid layouts and navigational zones reused to enforce structural consistency.

INTERFACE PROTOTYPING

Progressive mockups that validate visual layout, interaction flows, and user requirements.



Storyboards

Paper sketches of screens and transitions — rapid and cheap

HTML Prototypes

Clickable web wireframes with active links but static data

Language Prototypes

Built in target programming language — near-final realism

Low fidelity · quick, low cost, disposable — High fidelity · realistic, measurable, near-production

Interface Evaluation

Four evaluation methodologies with progressively deeper user involvement, realism, and analytical rigor.

01

Heuristic Evaluation

Usability experts inspect the interface against established design heuristics (e.g. Nielsen's principles).

USER INVOLVEMENT

None — experts only

Strength: Catches 70%+ of major design flaws early and cheaply

02

Walk-through Evaluation

The project team walks users through the prototype, explaining workflows and gathering feedback.

USER INVOLVEMENT

Users watch & discuss

Strength: Validates conceptual understanding without working code

03

Interactive Evaluation

Users work directly with the prototype on assigned tasks while an analyst observes hesitations and errors.

USER INVOLVEMENT

Users try hands-on

Strength: Reveals unexpected usability bottlenecks in real time

04

Formal Usability Testing

Controlled laboratory experiment recording time-on-task, completion rates, and user interactions.

USER INVOLVEMENT

Users measured rigorously

Strength: Generates objective, quantitative usability benchmark data

Quicker · lower cost · early cycle → More realistic · empirical metrics · high investment

Navigation Design

Guiding principles and primary control mechanisms that ensure seamless, reversible user movement.

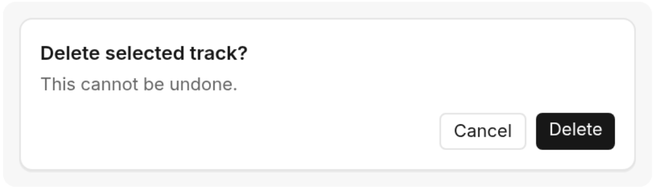
CORE NAVIGATION PRINCIPLES

Safety & Predictability

01 — ERROR DEFENSE

Prevent Mistakes

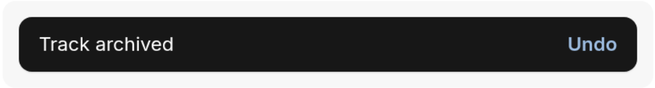
Label commands clearly, disable invalid choices, and confirm destructive actions.



02 — RESILIENCE

Simplify Recovery

Make actions reversible; keep Undo, Cancel, and Back visible and immediate.



03 — CONSISTENCY

Consistent Grammar

Standardize command syntax uniformly: either object-then-action or action-then-object.

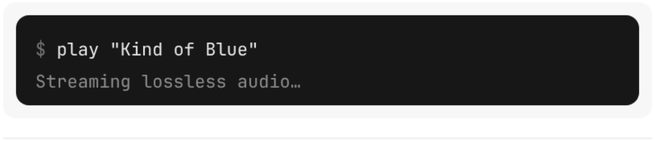


NAVIGATION CONTROLS

Interaction Paradigms

Language Commands

Users enter textual commands or natural-language syntax.



Pro: Fast for experts

Con: Requires recall

Menus

Users choose from a categorized visual list of options.



Pro: Zero memorization

Con: Slower for power users

Direct Manipulation

Users interact with visual objects directly (drag, drop, resize).



Pro: Natural tactile feedback

Con: Hard for bulk tasks

Navigation Components & Menu Types

Comparative taxonomy of all major navigation controls, component structures, and ideal deployment scenarios.

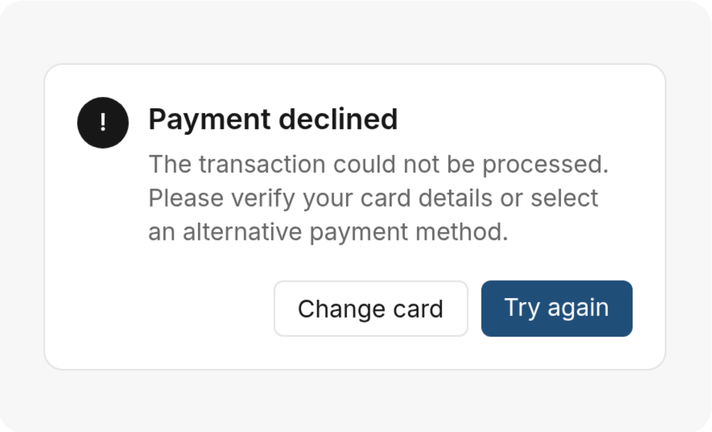
COMPONENT / CONTROL	CATEGORY	STRUCTURAL MECHANISM	BEST USE CASE / RECOMMENDATION
Menu Bar	Global	Horizontal command strip pinned across the screen top	Primary entryway for major top-level system modules
Drop-down Menu	Hierarchical	Cascading list revealing options beneath a parent item	Grouping second-level commands without cluttering the screen
Hyperlink Menu	Page Index	Structured list of text links to destination pages	Main navigational directory for web portals and sidebars
Embedded Hyperlink	In-Content	Clickable inline text links woven into paragraph copy	Contextual references to related records or definitions
Pop-up / Context Menu	Contextual	Floats at cursor coordinates on right-click or long-press	Power-user shortcuts tailored to the selected object
Tab Menu / Tabs	In-Page Views	Notebook-style tab headers switching between parallel views	Switching related forms or views without leaving the screen
Breadcrumbs	Hierarchy	Linear parent-child trail (Home › Category › Item)	Deep taxonomies to show exact path and easy return route
Search Bar	Direct Access	Query text input with live filtering and auto-suggestions	Fast keyword navigation that bypasses deep browsing
Image Map	Spatial	Clickable hotspot coordinates defined over a visual graphic	Spatial contexts where the image carries meaning (maps, seating)

Selection Rule: Match component visibility to usage frequency — global structures stay pinned, contextual controls appear on demand.

10 Navigation Controls Compared

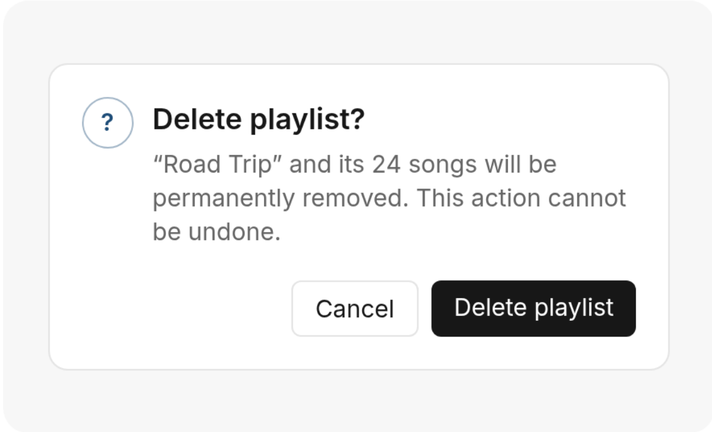
Message Design

Effective system feedback guides user action, prevents data loss, and preserves user confidence.



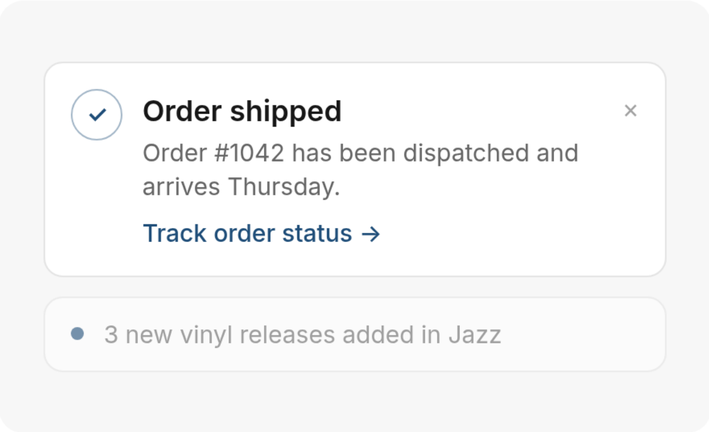
Error Messages

State clearly what happened and how to fix it in constructive terms — never blame the user or display cryptic machine codes.



Confirmation Messages

Prompt only before irreversible or destructive actions; explicitly name the action on the confirmation button.



Notification Messages

Acknowledge system state changes or background completions non-intrusively without blocking active work.

GOLDEN RULES Be clear, concise, polite, and actionable — speak in user-oriented vocabulary, never raw system errors.

HCI Feedback Standard

Input Design Overview

Capturing accurate information at the source through human-centered design principles and automated hardware capture.

CORE INPUT DESIGN PRINCIPLES

Accuracy & Integrity

01
Data Accuracy
Capture clean, error-free data on first entry to prevent corruption.

02
Data Integrity
Enforce business rules, schema constraints, and valid references.

03
Keystroke Reduction
Use smart defaults, structured selection lists, and auto-complete.

04
Source Data Automation
Capture data electronically where it originates to eliminate re-keying.

DATA CAPTURE TECHNOLOGIES (SOURCE DATA AUTOMATION)

Hardware & Electronic Entry

Barcode Reader
Scans optical codes instantly

Used in: Retail POS & warehouse inventory

OCR
Converts printed text into digital data

Used in: Paper forms, cheques & archives

Smart Card
Microchip stores & authenticates securely

Used in: EMV bank cards & national IDs

RFID
Wireless tags read without line-of-sight

Used in: Toll systems & logistics tracking

Web Forms
Captures structured input directly online

Used in: E-commerce & customer portals

Online vs Batch Processing

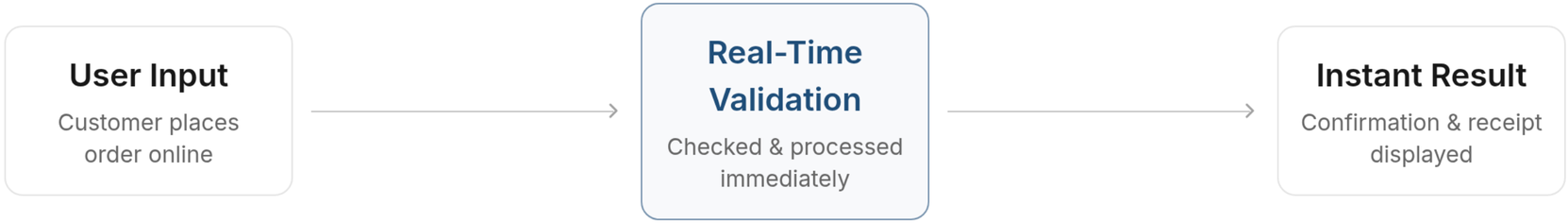
Architectural comparison between real-time transaction handling and queued batch execution.

Online Processing

Immediate

Each transaction is processed, validated, and persisted the moment it is submitted.

e.g. ATM cash withdrawal, flight seat reservation, e-commerce checkout

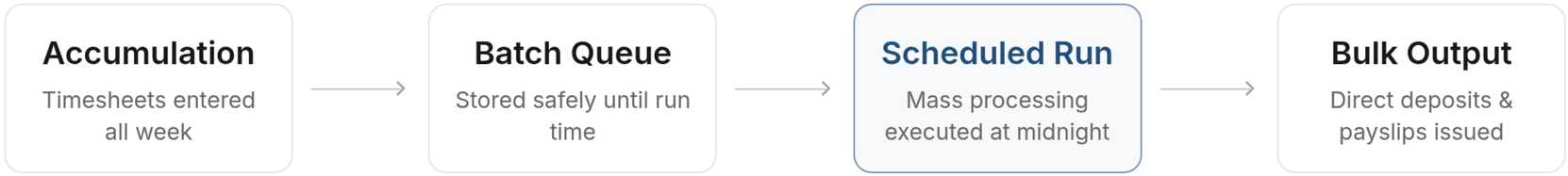


Batch Processing

Scheduled

Transactions are gathered over an operational period and processed together at scheduled times.

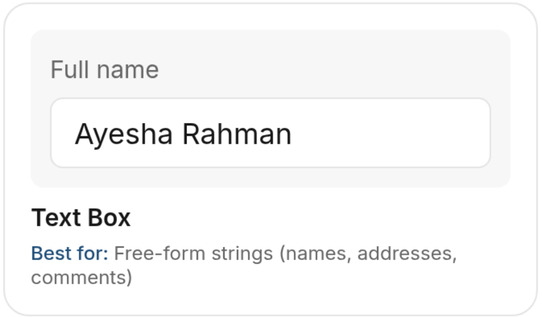
e.g. Corporate payroll, monthly utility billing, end-of-day bank clearing



Core Design Recommendation: Prefer online processing whenever possible because errors are flagged instantly while the user is actively present to rectify them.

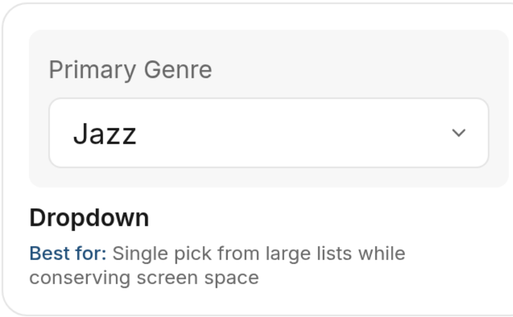
Input Controls

Selecting the optimal form control to minimize typing, enforce data integrity, and accelerate user entry.



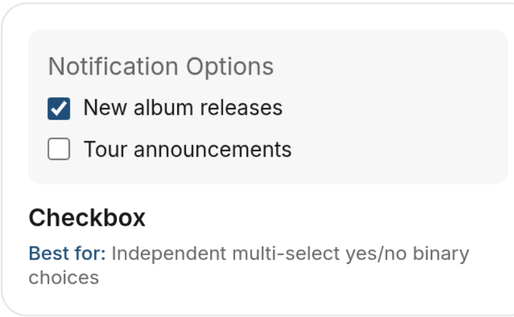
Text Box

Best for: Free-form strings (names, addresses, comments)



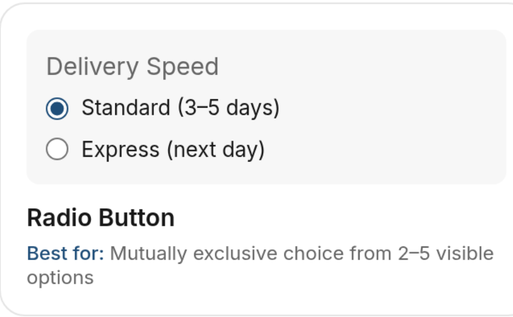
Dropdown

Best for: Single pick from large lists while conserving screen space



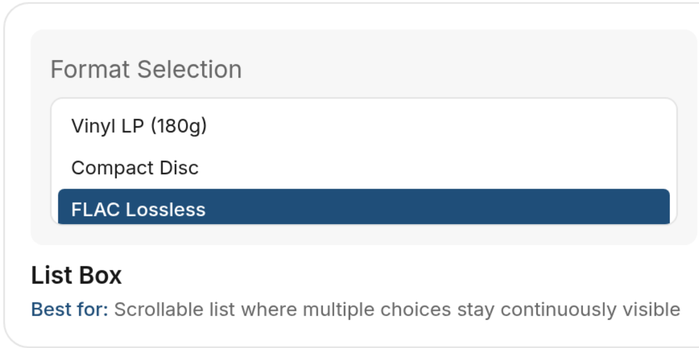
Checkbox

Best for: Independent multi-select yes/no binary choices



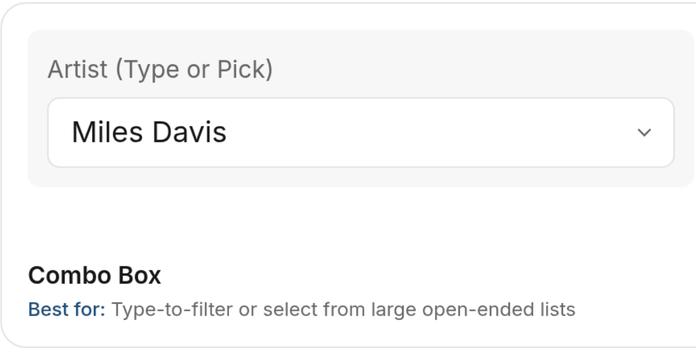
Radio Button

Best for: Mutually exclusive choice from 2–5 visible options



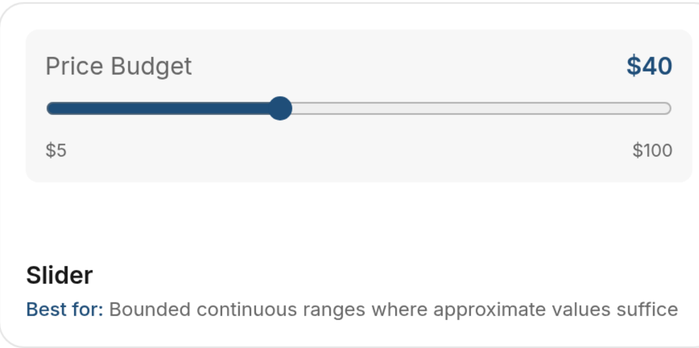
List Box

Best for: Scrollable list where multiple choices stay continuously visible



Combo Box

Best for: Type-to-filter or select from large open-ended lists



Slider

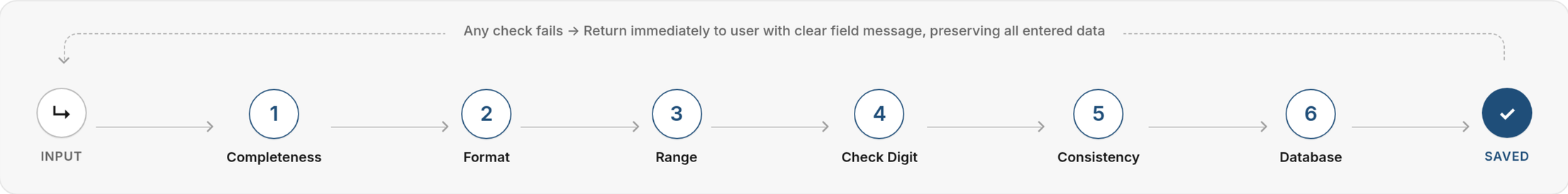
Best for: Bounded continuous ranges where approximate values suffice

Input Validation

Multi-tier validation pipeline ensuring data accuracy, integrity, and safety before database persistence.

VALIDATION PIPELINE

Sequential Verification Stages



THE SIX ESSENTIAL VALIDATION CHECKS

Comprehensive Audit Criteria

Check 1
Completeness Check
All required fields present

"Full Name is required"

Check 2
Format Check
Matches expected type & syntax

"Must be a valid email (@)"

Check 3
Range Check
Values within logical bounds

"Quantity must be 1–10"

Check 4
Check Digit Check
Mathematical algorithm check

"Luhn checksum verified"

Check 5
Consistency Check
Interdependent fields agree

"End date after start date"

Check 6
Database Check
Referential record integrity

"SKU in inventory & in stock"

Output Design & Tune Source Case Study

Foundational output principles synthesized with an end-to-end customer journey demonstrating how all lecture domains unite.

OUTPUT DESIGN FOUNDATIONS

Principles · Types · Formats

Output Principles

Core Quality

Relevance
Targeted data needed for immediate decisions

Accuracy
Verifiable, correct and unbiased figures

Timeliness
Delivered in time to take meaningful action

Accessibility
Clear across devices, screen sizes & abilities

Output Types

Presentation

Reports: Structured detail, summary & exception documents

Dashboards: Real-time operational KPIs & status monitoring

Notifications: Actionable alerts via push, SMS, email & in-app

Output Formats

Delivery Media

Print	Web	Mobile	PDF	Excel
Archival records	Interactive views	On-the-go alerts	Fixed invoices	Data analysis

TUNE SOURCE CASE STUDY · END-TO-END USER JOURNEY

How All 5 Disciplines Work Together

1 → 2 → 3 → 4 → 5

UI & NAVIGATION
Browse Music
Filter by genre or search artist
System: Catalog grid, search & breadcrumbs

UI & INPUT CONTROLS
View Album
Audition clip & pick format
System: Audio player & format dropdown

NAVIGATION & FEEDBACK
Add To Cart
Confirm quantity & options
System: Cart badge & toast confirmation

INPUT & VALIDATION
Checkout
Enter shipping & payment
System: Autofill & Luhn card validation

OUTPUT DESIGN
Confirmation
Review invoice & track order
System: Dashboard receipt & PDF invoice

Architectural Synergy: UI Design attracts & orientates → Navigation guides → Input & Validation capture error-free data → Output confirms & delivers value.

Tune Source Music Store System

Interactive Seminar Activity

Q SEMINAR DISCUSSION PROMPT

How does consistency improve user experience?

Analyze the compounding impact of internal and external consistency on user confidence, learning curves, and system errors.

Cognitive Load Reduction
Users transfer mental models effortlessly without pausing or relearning.

Error Prevention & Safety
Standardized buttons and confirmations prevent accidental misclicks.

Speed & Muscle Memory
Predictable workflows and shortcuts maximize user throughput.

Design System Governance
Shared tokens, templates, and audits maintain coherence at scale.

LECTURE SUMMARY · CORE CURRICULUM PILLARS

Complete Topic Synthesis

01 PILLAR 1
UI Principles
Layout, aesthetics, UX, consistency & the 3-clicks rule.

02 PILLAR 2
UI Process
Iterative cycle from use scenarios to evaluation.

03 PILLAR 3
Navigation Design
Clear hierarchies, mistake prevention & reversible routes.

04 PILLAR 4
Input Design
Source data capture, optimal controls & typing reduction.

05 PILLAR 5
Validation
Multi-tier verification pipeline before data commit.

06 PILLAR 6
Output Design
Relevant, timely reports & dashboards across media.

Thank You.

Questions & Academic Discussion

System Analysis & Design
Department of Computer Science & Engineering

● Course Lecture Presentation Complete · 20 Slides